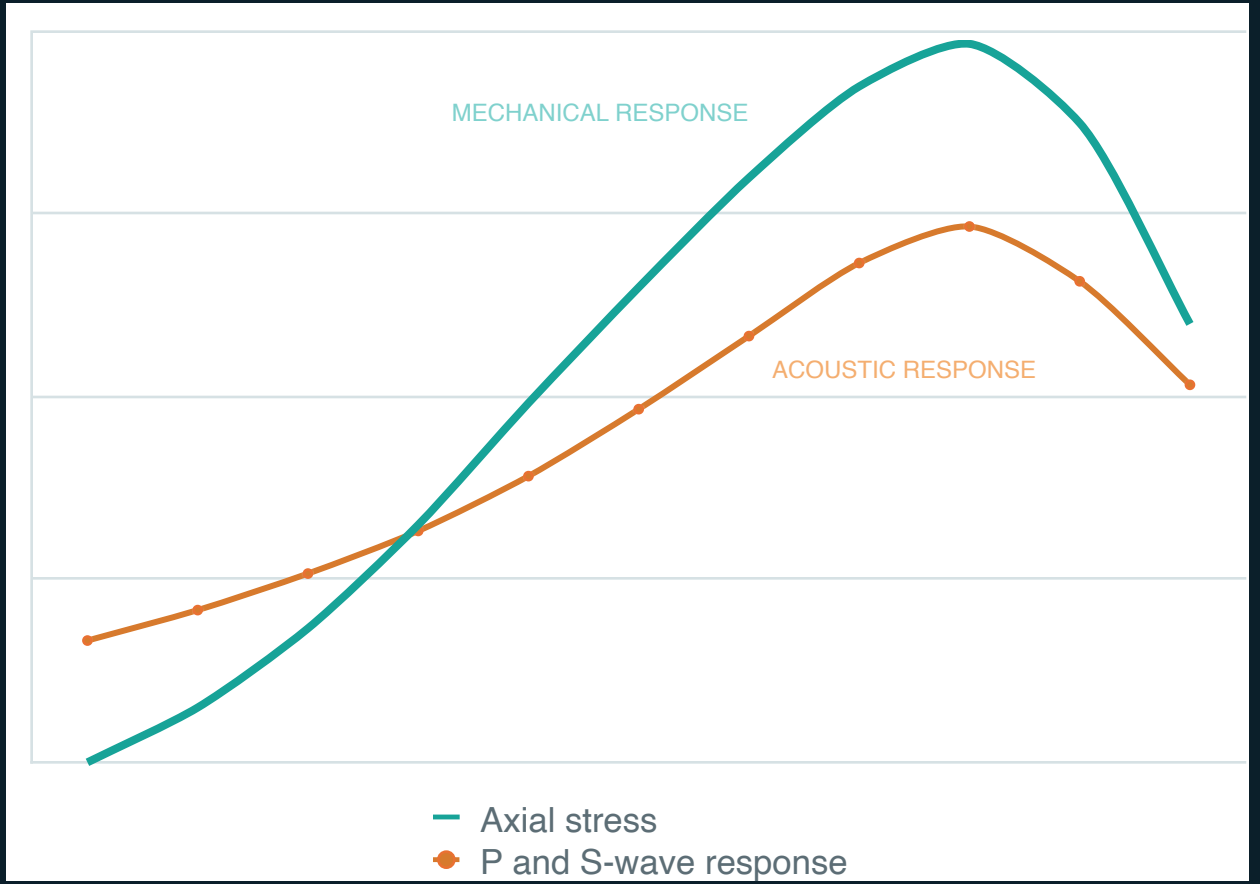
LECTURE 7

UCS data analysis

From recorded signals to a comparison of rock behavior



Mechanical question

How does each specimen deform, stiffen and fail?

Acoustic question

How do P and S arrivals evolve during loading?

Goal: connect both observations before comparing rock types

Analysis workflow and file handoffs

1

Mechanical reduction

TDMS file

acousto_mech.txt

2

Waveform analysis

Waveform TSV folder

Synchronized waveform picks (CSV)

3

One experiment

Two synchronized outputs

Stress, velocity and Vp/Vs

4

Multiple experiments

Comparable rock tests

Overlaid mechanical and acoustic curves

Check units, specimen geometry, system delays, synchronization and accepted picks before comparing curves.

Mechanical data reduction

1

Inspect the TDMS record

Metadata, PID settings, comments and every ADC channel

Output retains every ADC channel, including AUX 5 for waveform synchronization.

What is a TDMS file?

A hierarchical binary container that stores measured arrays and the metadata needed to interpret them



One experiment file

- Fast binary storage
- Many synchronized channels
- Properties at file, group and channel level

Core hierarchy

FILE › GROUP › CHANNEL

Structure of the UCS experiment file

Group and channel names are the example used in this course

TDMS FILE
triax_try_granit.tdms

File properties
Date, operator, starting PID settings

GROUP: ADC
Recorded time series

- **CHANNEL: Time** — one value for every ADC record
- **CHANNEL: Vertical Load** — raw array + slope + intercept + unit
- **CHANNEL: Vertical Displacement** — raw array + slope + intercept + unit
- **CHANNEL: AUX 5** — voltage used to synchronize waveforms

GROUP: Metadata
Events during acquisition

- **Comment Time + Comment** — experiment notes mapped to the nearest ADC row

Other groups may store additional hardware or acquisition information

Inside one channel

Example: Vertical Load

Numeric array

```
[0.012, 0.014, 0.019, …]
```

Channel properties

```
Slope m
Intercept b
Unit kN
```

x_cal = m x_raw + b
calibrated series in kN

Ready for stress calculation

Key idea: a TDMS file is a tree of measurements and properties, not a flat spreadsheet.

Mechanical data reduction

1 **Inspect the TDMS record**

Metadata, PID settings, comments and every ADC channel

2 **Calibrate the measurements**

xcal = m xraw + b; retain raw and calibrated channels

3 **Correct electrical steps**

Align the end row to the start row; shift all following records

4 **Calculate mechanical variables**

Zero force and displacement, then calculate stress and strain

SCHEMATIC CORRECTION



Engineering quantities

$A_0 = \pi D_0^2 / 4$ $\sigma_1 = (F - F_0) / A_0$

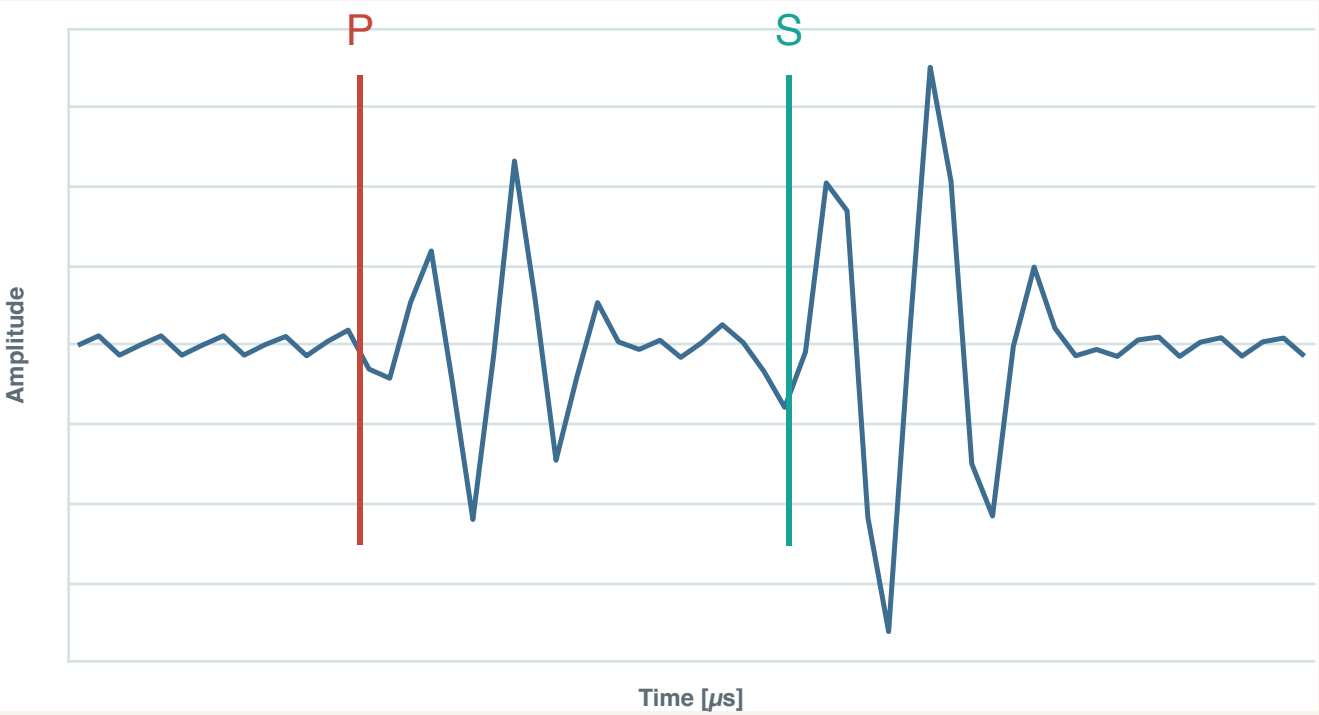
$\Delta L_corrected = \Delta L_recorded - (F - F_0)/k_m$

$\varepsilon_a = \Delta L_corrected / L_0$

Output retains every ADC channel, including AUX 5 for waveform synchronization.

Ultrasonic waveform analysis

ONE TRACE



Arrival time is the first repeatable departure from the baseline, not the largest peak.

1 **Link files to synchronization intervals**

Mechanical sync edges pair each waveform file with the loading record.

2 **Inspect all traces**

Use the overview to find representative traces and identify poor signal quality.

3 **Pick P and S arrivals**

Manual, STA/LTA or cross-correlation; apply one convention consistently.

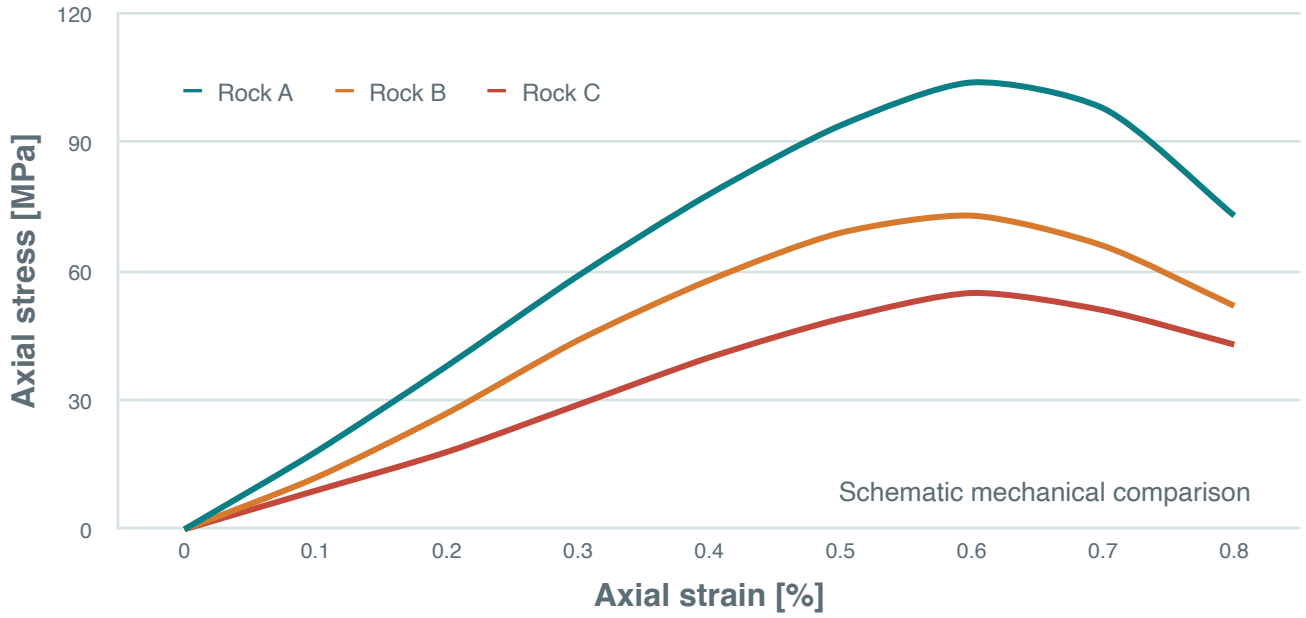
4 **Validate before export**

Check stacked picks for missing points, sudden jumps and cycle skips.

Output: synched_waveform_picks.csv with mechanical time, row index and all picking methods

Integration and comparison across rock types

MULTIPLE EXPERIMENTS



Comparison rule Use shared variables and units. Keep the picking method and quality threshold consistent whenever possible.

Final interpretation combines strength and strain evolution with P velocity, S velocity and Vp/Vs for each rock type.

Today we build the chain from raw records to a defensible comparison.

Feedback welcome

